

# Practical No.11

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## Objective:

### Create a dataset of student's marks and attendance

1. Find the data type of all the columns.
2. Display the summary statistics for all numeric columns.
3. Display the range for all numeric columns.
4. Compute average score for each student.
5. Add Remarks column by writing 'Excellent', 'Good' and 'Needs improvement' for students having marks greater than 90, 82 and less than 82 respectively.
6. Find the correlation between Maths and Science marks.
7. Find the Variance and Covariance matrix.

## Python Codes:

### Creation of Dataframe

```
In [1]: import pandas as pd
import numpy as np

# Create the DataFrame
data = {
    'Student': ['Aisha', 'Bilal', 'Charan', 'Deepa', 'Fatima', 'Gaurav', 'Harini', 'Imran', 'John', 'Kavya'],
    'Maths': [88, 92, 79, 85, 90, 76, 95, 89, 84, 91],
    'Science': [82, 85, 78, 80, 83, 88, 90, 86, 84, 89],
    'English': [75, 80, 70, 82, 78, 85, 88, 80, 79, 87],
    'Attendance(%)': [92, 95, 88, 90, 96, 85, 98, 93, 89, 97]
}

df = pd.DataFrame(data)
df
```

```
Out[1]:
```

|   | Student | Maths | Science | English | Attendance(%) |
|---|---------|-------|---------|---------|---------------|
| 0 | Aisha   | 88    | 82      | 75      | 92            |
| 1 | Bilal   | 92    | 85      | 80      | 95            |
| 2 | Charan  | 79    | 78      | 70      | 88            |
| 3 | Deepa   | 85    | 80      | 82      | 90            |
| 4 | Fatima  | 90    | 83      | 78      | 96            |
| 5 | Gaurav  | 76    | 88      | 85      | 85            |
| 6 | Harini  | 95    | 90      | 88      | 98            |
| 7 | Imran   | 89    | 86      | 80      | 93            |
| 8 | John    | 84    | 84      | 79      | 89            |
| 9 | Kavya   | 91    | 89      | 87      | 97            |

### 1. Data type of all the columns

```
In [2]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   Student         10 non-null    object
1   Maths           10 non-null    int64
2   Science         10 non-null    int64
3   English         10 non-null    int64
4   Attendance(%)   10 non-null    int64
dtypes: int64(4), object(1)
memory usage: 532.0+ bytes

```

## 2. Display the summary statistics for all numeric columns.

In [3]: `df.describe()`

Out[3]:

|              | Maths     | Science   | English   | Attendance(%) |
|--------------|-----------|-----------|-----------|---------------|
| <b>count</b> | 10.000000 | 10.000000 | 10.000000 | 10.000000     |
| <b>mean</b>  | 86.900000 | 84.500000 | 80.400000 | 92.300000     |
| <b>std</b>   | 5.933895  | 3.894444  | 5.481281  | 4.270051      |
| <b>min</b>   | 76.000000 | 78.000000 | 70.000000 | 85.000000     |
| <b>25%</b>   | 84.250000 | 82.250000 | 78.250000 | 89.250000     |
| <b>50%</b>   | 88.500000 | 84.500000 | 80.000000 | 92.500000     |
| <b>75%</b>   | 90.750000 | 87.500000 | 84.250000 | 95.750000     |
| <b>max</b>   | 95.000000 | 90.000000 | 88.000000 | 98.000000     |

## 3. Display the range for all numeric columns.

In [4]: `range_values = df[['Maths', 'Science', 'English']].max() - df[['Maths', 'Science', 'English']].min()`  
`range_values`

Out[4]:

|         |    |
|---------|----|
| Maths   | 19 |
| Science | 12 |
| English | 18 |

dtype: int64

## 4. Compute Average Score for each student

In [5]: `df['Average_Score'] = df[['Maths', 'Science', 'English']].mean(axis=1)`  
`df[['Student', 'Average_Score']]`

Out[5]:

|          | Student | Average_Score |
|----------|---------|---------------|
| <b>0</b> | Aisha   | 81.666667     |
| <b>1</b> | Bilal   | 85.666667     |
| <b>2</b> | Charan  | 75.666667     |
| <b>3</b> | Deepa   | 82.333333     |
| <b>4</b> | Fatima  | 83.666667     |
| <b>5</b> | Gaurav  | 83.000000     |
| <b>6</b> | Harini  | 91.000000     |
| <b>7</b> | Imran   | 85.000000     |
| <b>8</b> | John    | 82.333333     |
| <b>9</b> | Kavya   | 89.000000     |

## 5. Add Remarks Column

```
In [6]: df['Remark'] = np.where(
        df['Average_Score'] >= 90, 'Excellent',
        np.where(df['Average_Score'] >= 82, 'Good', 'Needs Improvement')
    )
df[['Student', 'Average_Score', 'Remark']]
```

```
Out[6]:
```

|   | Student | Average_Score | Remark            |
|---|---------|---------------|-------------------|
| 0 | Aisha   | 81.666667     | Needs Improvement |
| 1 | Bilal   | 85.666667     | Good              |
| 2 | Charan  | 75.666667     | Needs Improvement |
| 3 | Deepa   | 82.333333     | Good              |
| 4 | Fatima  | 83.666667     | Good              |
| 5 | Gaurav  | 83.000000     | Good              |
| 6 | Harini  | 91.000000     | Excellent         |
| 7 | Imran   | 85.000000     | Good              |
| 8 | John    | 82.333333     | Good              |
| 9 | Kavya   | 89.000000     | Good              |

## 6. Find Correlation between Maths and Science

```
In [7]: df[["Maths", "Science"]].corr()
```

```
Out[7]:
```

|         | Maths    | Science  |
|---------|----------|----------|
| Maths   | 1.000000 | 0.406284 |
| Science | 0.406284 | 1.000000 |

## 7. Find the variance and covariance matrix

```
In [8]: df.cov(numeric_only=True)
```

```
Out[8]:
```

|               | Maths     | Science   | English   | Attendance(%) | Average_Score |
|---------------|-----------|-----------|-----------|---------------|---------------|
| Maths         | 35.211111 | 9.388889  | 11.711111 | 24.366667     | 18.770370     |
| Science       | 9.388889  | 15.166667 | 18.333333 | 7.166667      | 14.296296     |
| English       | 11.711111 | 18.333333 | 30.044444 | 8.644444      | 20.029630     |
| Attendance(%) | 24.366667 | 7.166667  | 8.644444  | 18.233333     | 13.392593     |
| Average_Score | 18.770370 | 14.296296 | 20.029630 | 13.392593     | 17.698765     |

## Interpretation:

1. The Column 'Student' is type of 'object', the column 'Maths' , 'Science' , 'English' and 'Attendance(%)' are type of ,int64' .
2. Summary statistics for all numeric columns are above .
3. the range for all numeric columns coressponding Maths,Science and English are 19,12 and 18.
4. Average Score for each student
5. Added a new column 'Remark' as their respective average score.
6. Correlation between Maths and Science' is 0.406284 which is positive correlation it mean if a student score in math then it he/she will also score in science .
7. variance and covariance matrix is computed and here is some key feature that. diagonal values are variances and off diagonal are covariances.

### variance of

Math, Science, English, Attendance(%) and Average\_Score Are 35.211111, 15.166667, 30.044444, 18.233333, and 17.698765 respectively

## similarly covariance between

(Math, Science) , (Math, English),(Math, Attendance(%) and (Math, Average\_Score) are 9.388889, 11.711111, 24.366667 and 18.770370 respectively

(Science, English), (Science, Attendance(%)) and (Science, Average\_Score) are 18.333333, 7.166667 and 14.296296 respectively

(English, Attendance(%)) and (English, Average\_Score) are 8.644444 and 20.029630 respectively

(Attendance(%), Average\_Score) is 13.392593.